CS 6320 — Assignment 6

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Semester: Summer 2026
Course: Deep Learning (CS 6320)

Part A — Portfolio tabular representation and model comparison (BGG)

Work type: Native tabular portfolio data (not proxy, not fallback case study).

Portfolio problem and representation

Item	Lock
Stakeholder	Hobby retailer screening preorder / initial stock for BGG-aware customers
Dataset	Kaggle *Board Games Database from BoardGameGeek* (`games.csv`, 21,925 rated games)
Unit	One row per game (`BGGId`)
Target	`high_rating = 1` if `AvgRating >= 7.0`, else `0` (~26.9% positive)
Representation	**Native tabular** — numeric design metadata, `GameWeight`, category flags, simple text-length features, parsed `goodplayers_count`
Part A path	Portfolio table directly (no Week 6 proxy translator)
Seed	`6320`
Implementation	`6320-hw6/scripts/`; end-to-end: `bash run_local.sh`

Features follow the Assignment 4 charter and Assignment 5 manifest: rating fields, rank columns, popularity counts, and mis-timed demand fields (`NumOwned`, `NumWant`, `NumWish`) are **hard-excluded**. Raw `Name` / `Description` text are not fed to models — only length/count proxies (same v1 charter).

Preprocessing and prediction-time rules

1. **Split first** — stratified random 70% / 15% / 15% by game row (unchanged from Assignment 5).
2. **Imputation** — numeric medians computed on **training rows only**, then applied to validation and test through `build_feature_matrix(..., numeric_medians=train_medians)`.
3. **Scaling** — `StandardScaler` fit on **training** features for logistic regression and MLP; trees use median-imputed raw features (scale-invariant).
4. **Categories** — eight `Cat:*` flags kept as 0/1 integers (low cardinality).
5. **Leakage** — no rating-derived or post-hoc demand columns in *X*; each game appears in one split only.

Embeddings: Not used. Category fields are binary flags, not repeated high-cardinality IDs. Numeric fields are not identity codes. *Embeddings were not appropriate because categories are low-cardinality one-hot flags and there are no stable high-cardinality entity IDs; I used scaled numerics plus binary category columns instead.*

Split and audit evidence

Split	Rows	Positive rate
train	15,347	26.9%
validation	3,289	27.7%
test	3,289	26.2%

Positive rates are stable across splits (within ~1.5 pp). Audit tables: `prep/bgg_split/split_audit/` (counts, numeric distributions, category positive rates).

Models compared (same target, split, metrics)

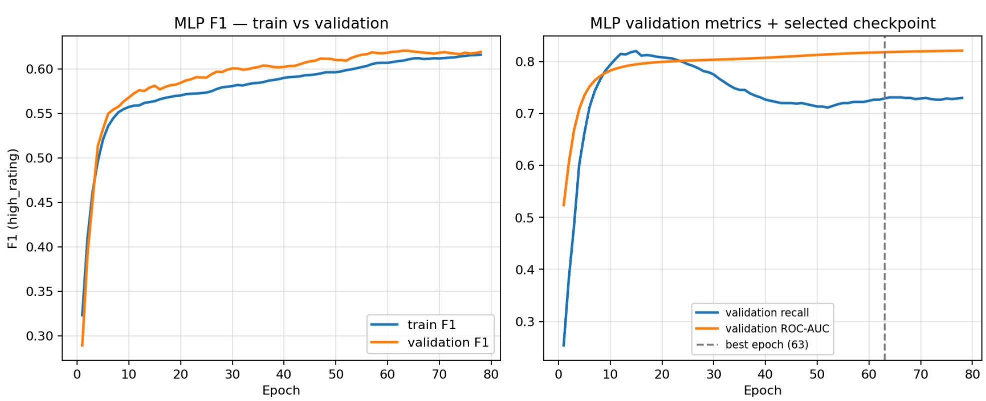
Model	Role	Key settings
Majority class	Sanity	Always predict not-high
Logistic regression	Simple linear baseline	L2, unweighted; train-fitted `StandardScaler`
Gradient boosted trees	Strong tabular benchmark	`HistGradientBoostingClassifier`, depth 6, lr 0.08, 300 trees
MLP	Neural tabular	64→32, ReLU, dropout 0.2, Adam (wd=1e-4), **BCE `pos_weight`** for imbalance, **early stop on validation F1** (patience 15; best epoch **63**)

Model selection uses **validation F1** on `high_rating=1` for the **MLP checkpoint** (early stop). Logistic and GBT use fixed configs; validation metrics guide comparison, and test metrics are reported once for final reporting. No hyperparameter search beyond the fixed configs above.

Results table

Model	Val F1	Val recall	Val ROC-AUC	Test F1	Test recall	Test ROC-AUC
Majority class	0.000	0.000	0.500	0.000	0.000	0.500
Logistic regression	0.527	0.422	0.815	0.544	0.454	0.832
Gradient boosted trees	**0.684**	0.623	**0.895**	**0.714**	0.660	**0.910**
MLP (early stop, epoch 63)	0.620	**0.729**	0.818	0.619	**0.747**	0.830

Validation precision: logistic **0.703**, GBT **0.759**, MLP **0.540**. Test accuracy: logistic **0.801**, GBT **0.861**, MLP **0.759**.



MLP training curves — validation checkpoint selection

Figure 1: MLP train/validation F1 and validation recall/ROC-AUC; dashed line = selected checkpoint (epoch 63). Validation recall peaks early (~epoch 15) then drifts down while F1 keeps improving — epoch 63 was chosen on validation F1, not peak recall.

Model-choice recommendation

Tabular modeling is justified for this portfolio problem: structured metadata and category flags carry signal well above chance (ROC-AUC ~0.91 test for GBT vs 0.50 majority).

Recommended staged model: gradient boosted trees, not the MLP.

- GBT wins on **validation and test F1** and **ROC-AUC** with the best precision–recall balance for a screening task.
- The MLP reaches **higher recall** (~0.75 test) but **lower F1 and precision** than GBT — it over-predicts positives, which wastes buyer attention (false stock flags).

- Logistic regression remains a useful **transparent floor** but under-recalls hits (~0.45 test recall), consistent with Assignment 5.
- Neural tabular complexity (checkpoint tuning, opaque weights, calibration risk from Assignment 5) is **not justified** when GBT beats the MLP on the primary ranking metrics under the same split.

Change trigger: If a time-based split collapses GBT advantage, or if recalibrated probabilities become a hard requirement, revisit logistic + Platt scaling before adding neural capacity.

Practical constraints

Constraint	Implication
Interpretability	Logistic and tree splits easier to explain to a buyer than MLP weights
Cost / ops	All models train in seconds on local CPU (~22k rows); GBT and MLP similar inference cost
Maintainability	Fixed sklearn/torch pipelines; manifest documents feature exclusions
Data size	Tabular row count is modest — no need for large-scale neural capacity
Monitoring	Track slice metrics (e.g. light vs heavy games from A5) if GBT is deployed internally

Responsible-use concern

BGG ratings reflect **enthusiast community taste**, not universal quality. Assignment 5 slices showed **weak performance on light games** (low positive rate, low recall). A tabular model — especially one tuned for recall like the MLP — can still **over-flag** niche titles. Outputs should remain **human-review screening aids**, not automated stock decisions; sensitive-proxy risk from title text is mitigated in v1 by using length features only, but franchise memorization should be audited before client-facing use.

Run evidence (local)

- Command: `bash run_local.sh` from repo root
- Data: `6320-hw2/part_b/data/bgg/games.csv` (or `GAMES_CSV`)
- Artifacts: `outputs/comparison_table.csv`, `outputs/*/summary.json`, `outputs/mlp_early_stop/mlp_history.csv`, `outputs/plots/mlp_training_curves.png`, `logs/run_local.log`
- Compute: local CPU only; CHPC not required

Part B — Portfolio checkpoint and model-choice note

Current data readiness

Usable. BGG `games.csv` is local, parsed, and split under the Assignment 4/5 charter (stratified hold-out, seed 6320). Feature manifest and split audits are reproducible via `prepare_bgg_data.py` and `run_split_audit.py`. No proxy or fallback path was needed.

Current baseline / model status

Stage	Status
Majority + logistic baseline	Complete (A5); logistic metrics reproduced in A6 comparison
Assignment 5 evaluation	Complete — imbalance, calibration gaps, slice weakness documented
Gradient boosted trees	**Trained (A6)** — current best tabular candidate
MLP tabular	**Trained (A6)** — competitive recall but secondary to GBT on F1/ROC-AUC
Time-based split	Not yet run — still open from charter
Text / image modeling	Not started — Week 7+ relevance TBD

Next planned experiment

Time-based split by `YearPublished` (recent-release validation cohort) with the **same feature manifest and GBT config**, plus **recalibration** (Platt or isotonic on validation, checked on test). This tests the charter's pre-release prediction story beyond random stratified hold-out.

Expected staged improvement

Honest pre-release metrics (likely lower than random split), better-calibrated probabilities for threshold decisions, and slice tables by release era. Success = documented trade-off between recall and precision under a decision-relevant split, not a single accuracy number.

How Week 6 evidence affects the final model-choice argument

Week 6 supports **keeping tabular methods central** to the portfolio: GBT on metadata alone reaches **test F1 0.71 / ROC-AUC 0.91**. It does **not** support elevating a neural tabular model — the MLP adds complexity without winning the comparison. The final presentation can argue: *structured features are sufficient for internal screening; modality-specific models (text embeddings, cover images) must beat this GBT protocol under a time split before they earn a place in the recommendation.*

Assignment 4 audit / charter updates

A4 item	Week 6 evidence	Update
Class imbalance	Logistic under-recalls; MLP boosts recall at precision cost; GBT balances best	**Confirmed** — metric choice must include F1/recall
Leakage exclusions	Same manifest; GBT gain is not from rating columns	**Reinforced**
GBT as initial candidate	Now trained; beats logistic and MLP on F1/AUC	**Success criterion met** for classical stage
Time-split generalization	Still random split only	**Still untested** — next experiment
Calibration / deployment	Not re-run in A6; GBT probabilities still need recalibration	**Still open**
Representativeness (light games)	Not re-sliced in A6	**Still flagged** from A5

Tabular relevance for the portfolio

Directly relevant. The client problem is inherently tabular metadata prediction. Embeddings are **not relevant** in v1 (low-cardinality categories). Simpler baselines remain **indirectly relevant** as floors and explainability anchors even if GBT becomes the primary candidate.

AI disclosure

Tool: Cursor (Composer)

AI-assisted: Repo scaffolding, model-comparison scripts, PyTorch MLP training loop, first-draft writeup prose.

My work: Locked native-tabular path; chose model set and fixed configs; ran and verified all metrics against saved outputs; model-choice recommendation based on comparison table.

Certification: I certify that all work not described above is my own, and I verified AI-generated content for accuracy.